

Karun Joseph E

Final year BTech ECE Student

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Bangalore, India

Electronics and Communication Engineering student at VIT with hands-on experience in embedded software, real-time systems, electronics, and hardware-software integration. Experienced with Embedded C, FreeRTOS, TI MCU+ SDK, SPI, UART, embedded debugging, peripheral drivers, sensor interfacing, and SoC-based development. Practical experience spans embedded platform development, mmWave radar interfacing, aerospace electronics, industrial automation, and electronics manufacturing. Focused on developing reliable embedded systems through low-level firmware, hardware integration, and systematic debugging.

Experience

VIT Vellore (TI-Sponsored Program)

May 2026 - Present

Embedded Software Intern

- Worked on embedded software and hardware-software integration for a heterogeneous multi-core SoC using TI MCU+ SDK and FreeRTOS.
- Developed and debugged SPI-based peripheral communication, including MCSPI and low-level driver interfaces, and interfaced an AWR2243 mmWave radar using mmWaveLink for device configuration and data acquisition.
- Used SysConfig, SDK examples, build systems, register-level debugging, GPIO/peripheral testing, serial logs, and UART-based CLI tools for platform integration, configuration, and firmware validation.

Team Sammart

April 2024 – August 2025

Electronics Engineer

- Gained hands-on experience across embedded electronics, PCB development, sensor integration, telemetry, RF communication, and hardware-software integration through real-world aerospace projects.
- Developed practical exposure to COTS flight computers, embedded avionics, hardware bring-up, debugging, testing, and subsystem integration across the electronics development lifecycle.
- Contributed to E-match manufacturing optimization for reliability, gaining experience in production-oriented hardware processes.

Nokia Solutions & Networks

June 2025 – July 2025

SMT & Industrial Engineering Intern

- Designed a Raspberry Pi-based ESD workstation monitoring system with real-time dashboard integration and contributed to an automatic UUT quality-verification project.

Flex Electronics

May 2025 – June 2025

Production Intern

- Performed functional testing of network routers and developed a Python-based SFP fault-detection dashboard to support troubleshooting.
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Education

Vellore Institute of Technology, Vellore

August 2023 - Present

B. Tech Electronics and Communications Engineering – **CGPA: 9.02**

Technical Skills

Programming: C, Embedded C, C++, Python, Verilog HDL

Embedded Systems: FreeRTOS, STM32, ESP32, TI J722s, Raspberry Pi, Embedded Debugging, Hardware-Software Integration

Interfaces & Protocols: SPI (MCSPI & MIBSPI), UART, I²C, GPIO, MIPI CSI-2

Hardware & Electronics: PCB Design, Prototyping, Sensor Integration, Hardware Bring-up, Data Acquisition, Telemetry, RF Communication, SMT/Mixed-Technology Soldering, Rework

Wireless & Radar: LoRa, XBee/ZigBee, BLE, mmWave Radar

Tools & Software: KiCad, Altium Designer, EasyEDA, VS Code, Git, Bitbucket, MATLAB, Simulink, CST Studio Suite, COMSOL Multiphysics, Cadence Virtuoso, TI SysConfig

Selected Projects

STM32 Flash Data Logger with Compression - Developed a flash-aware STM32 data-logging architecture using DPCM prediction, Zigzag transformation, and Rice entropy coding to compress high-rate sensor data before SPI flash storage. Achieved 84.1% space savings (6:1 compression) while reducing flash page usage from 63 to 10 pages; 3rd Place, SanDisk Hackathon 2026.

CanSat Flight Computer PCB & Avionics - Developed an STM32 based flight computer integrating GPS, sensors, XBee telemetry, payload control, and flight-data acquisition. Designed and integrated embedded electronics for reliable sensor acquisition, telemetry, and mission control.

Propulsion Testbench PCB for Solid Rocket Motors - Developed an ESP32-based embedded data-acquisition system integrating HX711 load-cell measurement, pressure sensing, strain sensing, SD-card data logging, LoRa telemetry, and remote ignition trigger. Integrated multiple hardware peripherals and developed firmware for real-time measurement, logging, and communication.

ESD Compliance Monitoring System - Raspberry Pi-based industrial ESD monitoring system with real-time dashboard.

Fuzzy Logic Obstacle Avoidance Robot - Designed and simulated a Mamdani fuzzy-logic obstacle-avoidance controller using MATLAB Fuzzy Logic Designer, then implemented the control algorithm on an ESP32 for real-time navigation.

PID-Controlled Gimbal Mechanism - Developed a servo-based gimbal stabilization system using PID control and real-time feedback for closed-loop positioning.

Soft Skills

Communication, Cross-Functional Collaboration, Problem-Solving, Adaptability. Project Coordination, Problem Solving